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Honours Analytics

ANN Assignment

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1 Linear Regression with Gibbs Sampling

a Derivations

Bayesian Linear Regression Derivation

$$\pi(\beta, \sigma^2) \propto \pi(\beta \mid \sigma^2)\, \pi(\sigma^2) \tag{1}$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta})\right) \left(\frac{1}{\sigma^2}\right)^{a-1} \exp\left(-\frac{b}{\sigma^2}\right) \tag{2}$$

Let $\tau = \frac{1}{\sigma^2}$:

$$\propto \exp\left(-\frac{\tau}{2}(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta})\right) \tau^{a-1} \exp(-b\tau) \tag{3}$$

$$\propto \tau^{a-1} \exp\left(-\frac{\tau}{2}(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta}) - b\tau\right) \tag{4}$$

Posterior

$$\pi(\beta, \sigma^2 \mid x, y) \propto \pi(\beta, \sigma^2)\, L(x, y; \beta, \sigma^2) \tag{5}$$

$$\propto \tau^{a-1} \exp\left(-\frac{\tau}{2}(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta}) - b\tau\right) \tag{6}$$

$$\times\, \tau^{n/2} \exp\left(-\frac{\tau}{2}(y - X\beta)'(y - X\beta)\right) \tag{7}$$

$$\propto \tau^{a+\frac{n}{2}-1} \exp\left[-\frac{\tau}{2}\left(SSR + (\beta - \hat{\beta})'(X'X)(\beta - \hat{\beta})\right.\right. \tag{8}$$

$$\left.\left.+ (\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta})\right) - b\tau\right] \tag{9}$$

Since $M^{-1} = M$:

$$= (\beta - \hat{\beta})'(X'X)(\beta - \hat{\beta}) + (\beta - \hat{\beta})' M (\beta - \hat{\beta}) \tag{10}$$

$$= (\beta - \hat{\beta})'[X'X + M](\beta - \hat{\beta}) \tag{11}$$

Marginal Posterior of τ

$$\pi(\tau \mid y, x) \propto \tau^{\frac{n}{2}+a-1} \exp\left(-\tau\left(\frac{SSR}{2} + b\right)\right) \tag{12}$$

$$\Rightarrow \tau \mid y, x \sim \text{Gamma}\left(a + \frac{n}{2},\, b + \frac{SSR}{2}\right) \tag{13}$$

Posterior of $\beta \mid \sigma^2$

$$\pi(\beta \mid \sigma^2, x, y) \propto \pi(\beta \mid \sigma^2)\, L(x, y; \beta, \sigma^2) \tag{14}$$

$$\propto \frac{1}{|\sigma^2 M|^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta})\right) \tag{15}$$

$$\times\, \frac{1}{(\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right) \tag{16}$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\left[(\beta - \hat{\beta})' M^{-1} (\beta - \hat{\beta}) + (y - X\beta)'(y - X\beta)\right]\right) \tag{17}$$

Completing the Square in Bayesian Linear Regression

Key Identity

$$(y - X\beta)'(y - X\beta) = SSR + (\beta - \hat{\beta})'(X'X)(\beta - \hat{\beta}) \tag{18}$$

Adding Prior Term

$$SSR + (\beta - \hat{\beta})'(X'X)(\beta - \hat{\beta}) + (\beta - \tilde{\beta})'M^{-1}(\beta - \tilde{\beta}) \tag{19}$$

Expand both quadratic terms:

$$= \beta'(X'X)\beta - 2\beta'(X'X)\hat{\beta} + \hat{\beta}'(X'X)\hat{\beta} \tag{20}$$

$$+ \beta'M^{-1}\beta - 2\beta'M^{-1}\tilde{\beta} + \tilde{\beta}'M^{-1}\tilde{\beta} + SSR \tag{21}$$

Group terms:

$$= \beta'(X'X + M^{-1})\beta - 2\beta'(X'X\hat{\beta} + M^{-1}\tilde{\beta}) + C \tag{22}$$

where C collects constants.

Complete the Square

Let

$$\Sigma^{-1} = X'X + M^{-1} \tag{23}$$

$$\mu = \Sigma\left(X'X\hat{\beta} + M^{-1}\tilde{\beta}\right) \tag{24}$$

Then:

$$= (\beta - \mu)'\Sigma^{-1}(\beta - \mu) + C \tag{25}$$

Useful Expansion Identity

$$(\beta - \mu)'\Sigma^{-1}(\beta - \mu) = \beta'\Sigma^{-1}\beta - 2\beta'\Sigma^{-1}\mu + \mu'\Sigma^{-1}\mu \tag{26}$$

Posterior Form

$$\pi(\beta \mid \sigma^2, y, X)\frac{1}{} \propto \exp\left(-\frac{1}{2\sigma^2}(\beta - \mu)'\Sigma^{-1}(\beta - \mu)\right) \tag{27}$$

$$\Rightarrow \beta \mid \sigma^2, y, X \sim \mathcal{N}(\mu, \sigma^2\Sigma) \tag{28}$$

Final Expressions

$$\Sigma = (X'X + M^{-1})^{-1} \tag{29}$$

$$\mu = \Sigma\left(X'X\hat{\beta} + M^{-1}\tilde{\beta}\right) \tag{30}$$

b Code for posterior samples

Creating Gibbs function

```
gibbs <- function(y,x,n_iter, M, beta_til, a, b){
  n = nrow(x)
  k = ncol(x)

  beta_til = as.matrix(beta_til)

  beta_samples = matrix(0, n_iter, k)
  var_samples = numeric(n_iter)

  #Starting values
  beta = rep(0, k)
  var = 1

  for (i in 1:n_iter){
    mu_beta <- solve(M + t(x)%*%x)%*%(t(x)%*%y + M%*%beta_til)

    # Sampling the betas
    beta <- MASS::mvrnorm(1, mu = mu_beta, Sigma = var * solve(M + t(x)%*%x))

    #Sample from sigmasqrd
    A2 <- t(y)%*%y + t(beta_til) %*% M %*% beta_til - t(mu_beta) %*% (M + t(x
       ) %*%x) %*% mu_beta

    shape = a + n/2
    rate = b + A2/2

    var = 1/rgamma(1, shape = shape, rate = rate)

    beta_samples[i,] = beta
    var_samples[i]  = var
  }
    return(list(beta = beta_samples, sigma2 = var_samples))
  }
```

c Analysis of 50 000 posterior samples

Trace Plots of Regression Coefficients

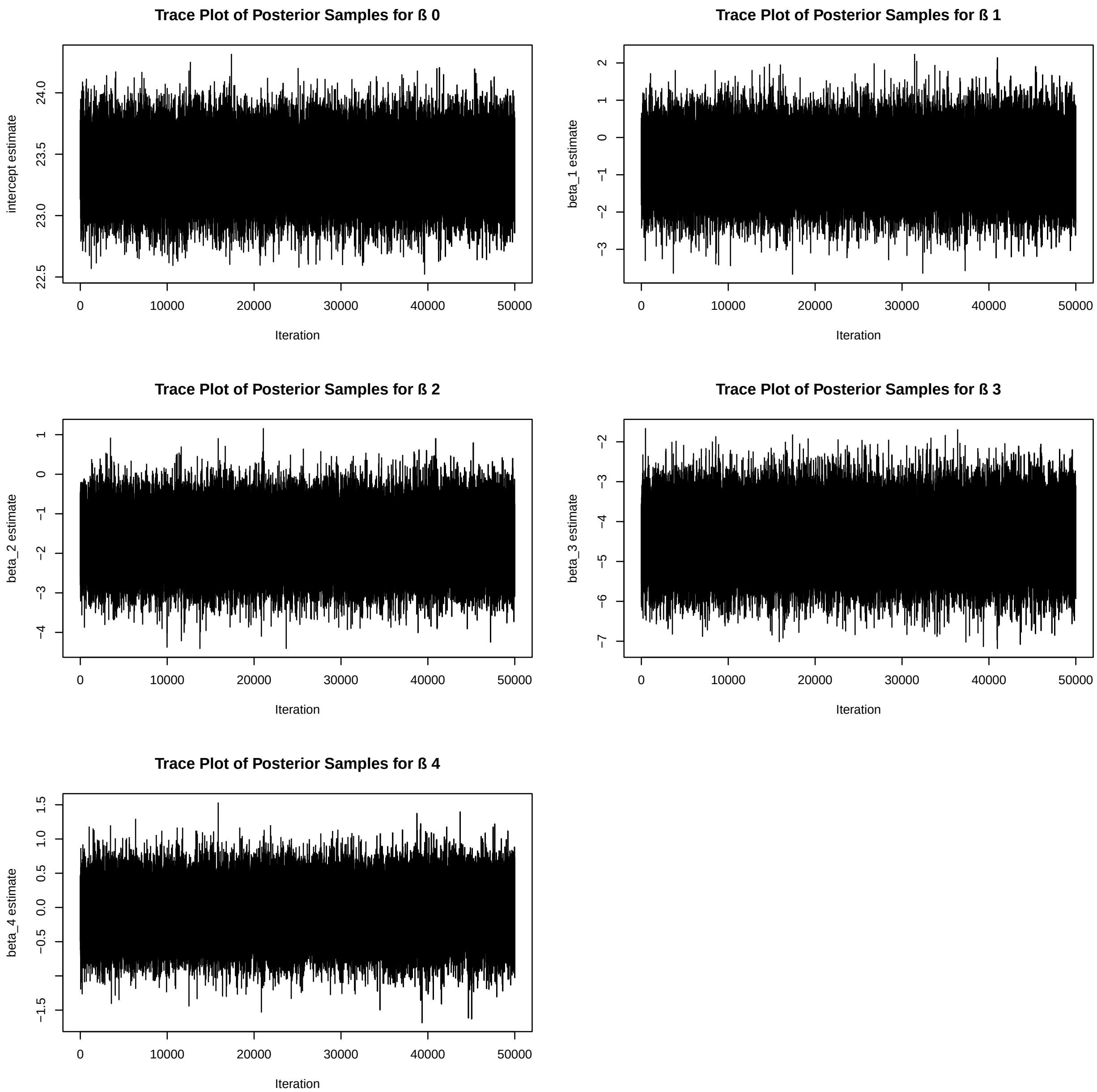

Figure 1: Trace plots of β_i estimates

The trace plots fluctuate around a stable mean and do not show any visible trends or drift over iterations. This suggests that the Markov chains have reached a stationary distribution. Therefore, we can conclude that the Gibbs sampler appears to have converged.

Posterior mean and OLS estimates

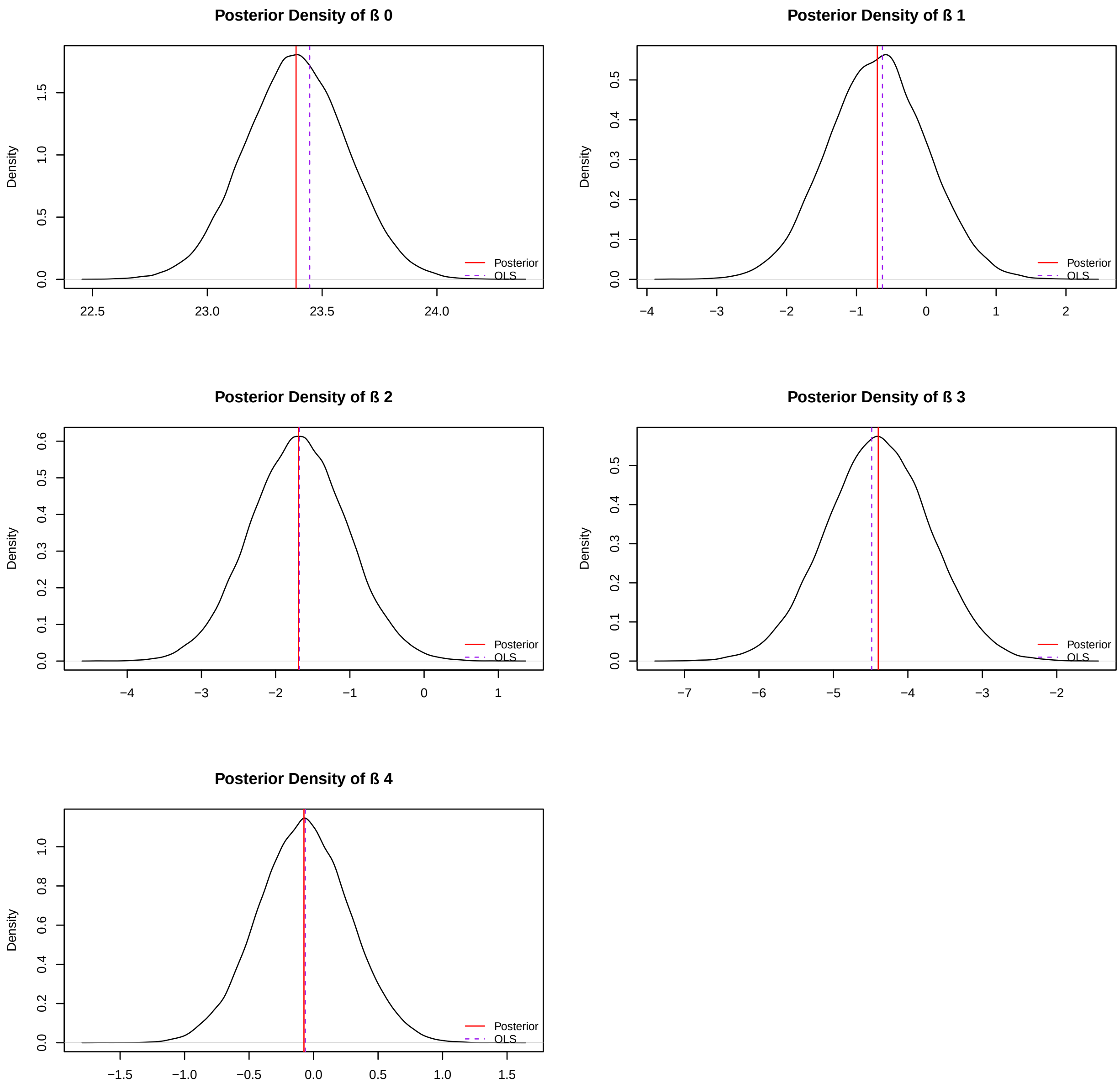

Figure 2: Density plots of β_i estimates

The plots in Figure 2 indicate that the posterior densities are approximately symmetric and unimodal. The posterior means closely match the OLS estimates, indicating that the likelihood is highly informative relative to the prior. The prior therefore has little influence on the posterior.

95% Credibility and Confidence Intervals

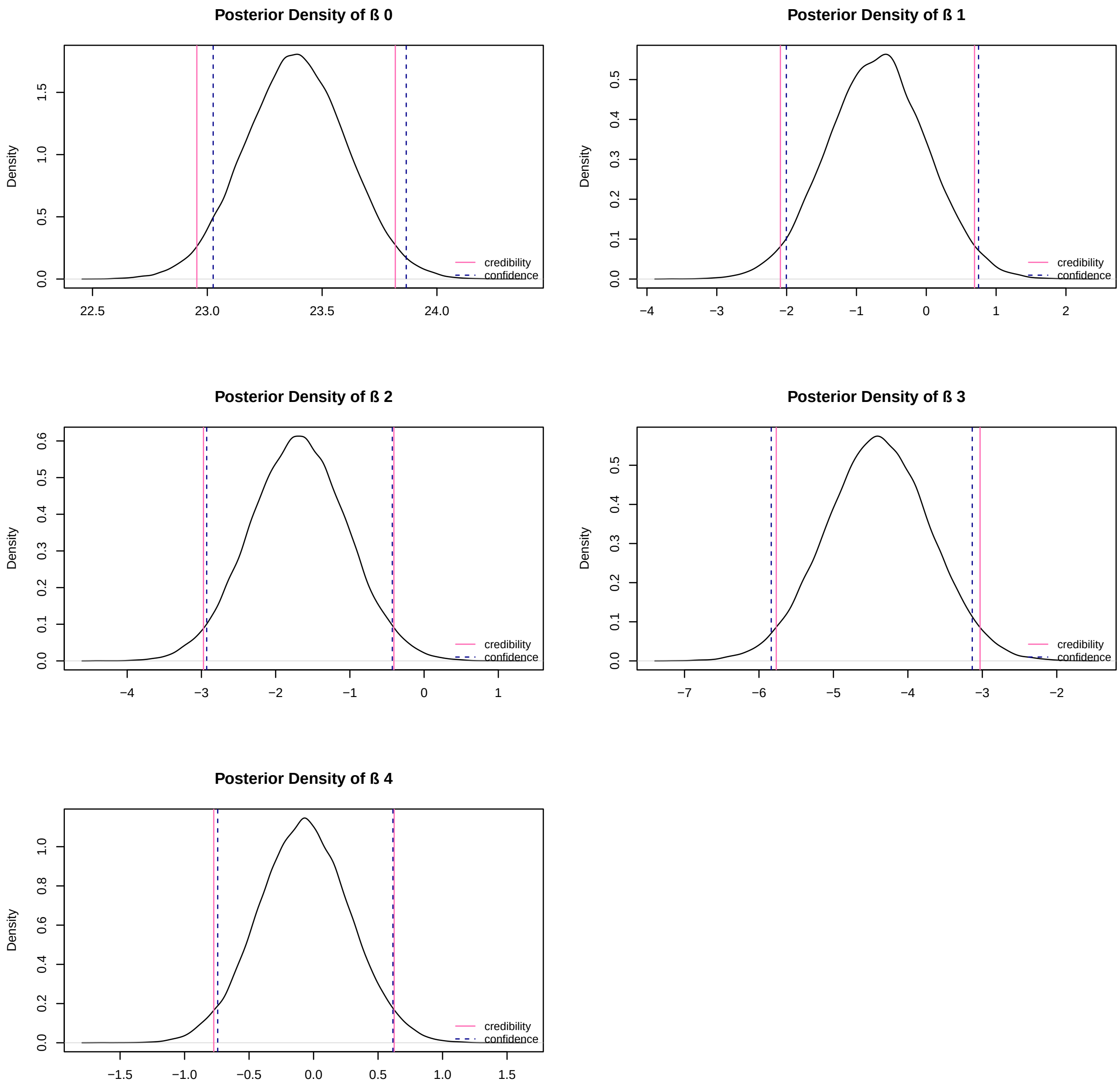

Figure 3: 95% Credibility and Confidence Intervals for β_i estimates

A 95% credibility interval means that there is a 95% probability that the true parameter value falls within the interval, given the observed data and prior. On the other hand, a 95% confidence interval means that if we computed the interval over many samples, 95% of those intervals would contain the true parameter value.

From Figure, the credibility and confidence intervals appear roughly similar across all parameter estimates. The greatest similarity is observed for the β_2 and β_4 estimates, where the confidence intervals are largely contained within the corresponding credibility intervals. The greatest disparity occurs for the intercept estimate, where the confidence interval is shifted to the right of the credibility interval. A similar, though smaller, shift to the right is observed for the β_1 estimate. For the β_3 estimate, the confidence interval is slightly to the left of the credibility interval. Overall, the general similarity between the credibility and confidence intervals further highlights that the prior is weak, and the posterior is largely driven by the data.

Predictor importance

The credibility intervals for the β_1 and β_4 estimates include 0, which suggests that displacement and acceleration may not be significant predictors of fuel efficiency. In contrast, the credibility intervals for the β_2 and β_3 estimates, corresponding to horsepower and weight, do not include 0, indicating that they are likely important predictors of fuel efficiency.

Some of these findings are unexpected. One would expected that a larger engine `displacement` would significantly reduce fuel efficiency, as larger engines usually consume more fuel. However, `horsepower` and `weight` being important predictors is expected, since heavier, more powerful vehicles would require more fuel, and are therefore less fuel efficient. The lack of significance of `acceleration` may be due to an overlap in information with `horsepower`, since both can measure vehicle performance.

d Posterior densities using an informative prior

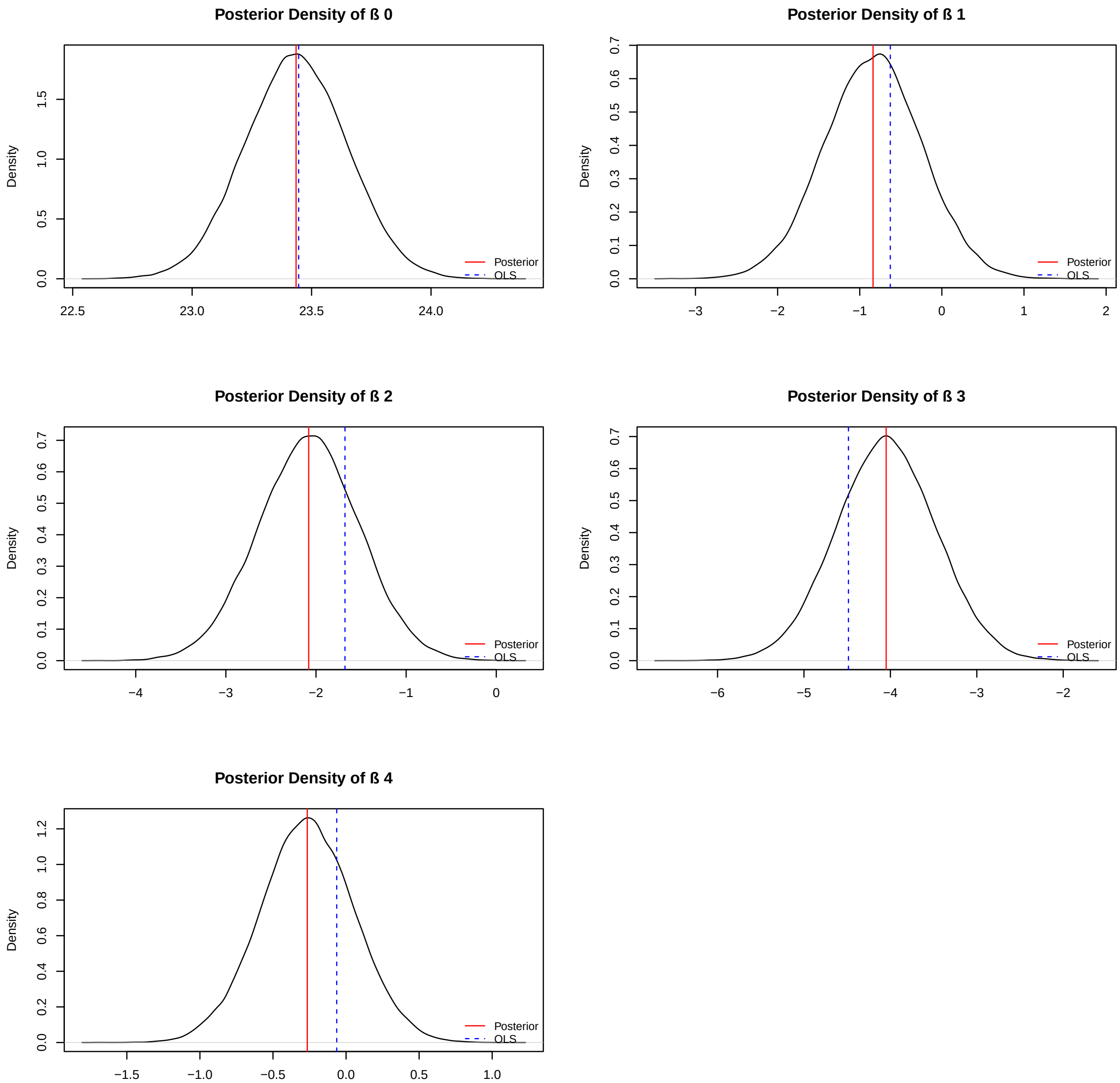

Figure 4: Posterior densities of β_i from informative prior

The posterior means and OLS estimates differ more under the informative prior than under the vague prior. For the intercept β_0, the OLS and posterior are almost identical under the informative prior, but showed larger disparity under the vague prior. In contrast, the remaining estimates show large differences between these values under the informative prior than the vague prior. This overall disparity suggests that the prior has a stronger influence on the posterior relative to the likelihood, leading to posterior estimates closer to the prior mean.

In general, the choice of prior affects how much influence is given to the prior relative to the data in the posterior. An informative prior holds a greater weight in determining the posterior, leading to larger disparities between OLS estimates and posterior means. A vague prior has little effect, causing the posterior to be largely driven by the data.

As the sample size increases, the data contains more information, resulting in a more informative likelihood. This increases the influence of the data relative to the prior in determining the posterior, causing the posterior mean to converge toward the OLS estimate, which is based only on the data.

Using only a subset of the data reduces the amount of information available in the likelihood, causing the posterior to be driven more by the prior distribution, which leads to a greater difference in the OLS estimates and posterior means.

2 A Bayesian Search

a Derivation of updating equations

We have been given the following variable definitions:

$$
\begin{aligned}
i &= \text{indices of grid cells} \\
Y_i &= \text{true state of whether fisherman is in } i^{th} \text{ cell} \\
\theta_i &= \text{probability of occurrence} = P(Y_i = 1) \\
Z_i &= \text{search result of } i^{th} \text{ cell} \\
p_i &= \text{probability of detection} = P(Z_i = 1 \mid Y_i = 1)
\end{aligned}
$$

We have also been given prior distributions

$$
\begin{aligned}
&Z_i \mid Y_i \sim \text{Bern}(Y_i p_i) = (Y_i p_i)^{z_i}(1 - Y_i p_i)^{1-z_i} \\
&Y_i \sim \text{Bern}(\theta_i) = \theta_i^{y_i}(1-\theta_i)^{1-y_i},
\end{aligned}
$$

and the following posterior distribution, where the i^{th} cell contains the fisherman but we fail to detect him

$$\pi(Y_i = 1 \mid Z_i = 0) = \frac{(1-p_i)\theta_i}{1 - p_i\theta_i}$$

Current square i:
We derive this as follows. By Bayes' theorem:

$$\pi(Y_i = 1 \mid Z_i = 0) = \frac{\pi(Z_i = 0 \mid Y_i = 1)\,\pi(Y_i = 1)}{\pi(Z_i = 0)}$$

Likelihoods:

$$
\begin{aligned}
&\pi(Z_i = 0 \mid Y_i = 0) = (0)^0(1-0)^1 = 1 \\
&\pi(Z_i = 0 \mid Y_i = 1) = (1 \cdot p_i)^0(1-p_i)^{1-0} = 1 - p_i
\end{aligned}
$$

Priors:

$$
\begin{aligned}
&\pi(Y_i = 1) = \theta_{i,\text{old}} \\
&\pi(Y_i = 0) = 1 - \theta_{i,\text{old}}
\end{aligned}
$$

Numerator:

$$\pi(Z_i = 0 \mid Y_i = 1)\,\pi(Y_i = 1) = (1 - p_i)\,\theta_{i,\text{old}}$$

Denominator:

$$
\begin{aligned}
\pi(Z_i = 0) &= \pi(Z_i = 0 \mid Y_i = 0)\,\pi(Y_i = 0) + \pi(Z_i = 0 \mid Y_i = 1)\,\pi(Y_i = 1) \\
&= 1 \cdot (1 - \theta_{i,\text{old}}) + (1 - p_i)\,\theta_{i,\text{old}} \\
&= 1 - \theta_{i,\text{old}} + \theta_{i,\text{old}} - p_i\theta_{i,\text{old}} \\
&= 1 - p_i\theta_{i,\text{old}}
\end{aligned}
$$

Therefore:

$$\theta_{i,\text{new}} = \pi(Y_i = 1 \mid Z_i = 0) = \frac{(1-p_i)\,\theta_{i,\text{old}}}{1 - p_i\theta_{i,\text{old}}}$$

Other squares $j \neq i$:
For squares not searched, we condition on $Z_i = 0$ being observed in square i:

$$\pi(Y_j = 1 \mid Z_i = 0) = \frac{\pi(Z_i = 0 \mid Y_j = 1)\,\pi(Y_j = 1)}{\pi(Z_i = 0)}$$

Numerator:

$\pi(Z_i = 0 \mid Y_j = 1) = 1$ (search is in i, so detection probability in j is 0)

$\pi(Y_j = 1) = \theta_{j,\text{old}}$

$$\pi(Z_i = 0 \mid Y_j = 1)\,\pi(Y_j = 1) = 1 \cdot \theta_{j,\text{old}} = \theta_{j,\text{old}}$$

Denominator (search i, fisherman not found):

$$\pi(Z_i = 0) = 1 - p_i\theta_{i,\text{old}}$$

Therefore:

$$\theta_{j,\text{new}} = \pi(Y_j = 1 \mid Z_i = 0) = \frac{\theta_{j,\text{old}}}{1 - p_i\theta_{i,\text{old}}}$$

b The probability heatmaps

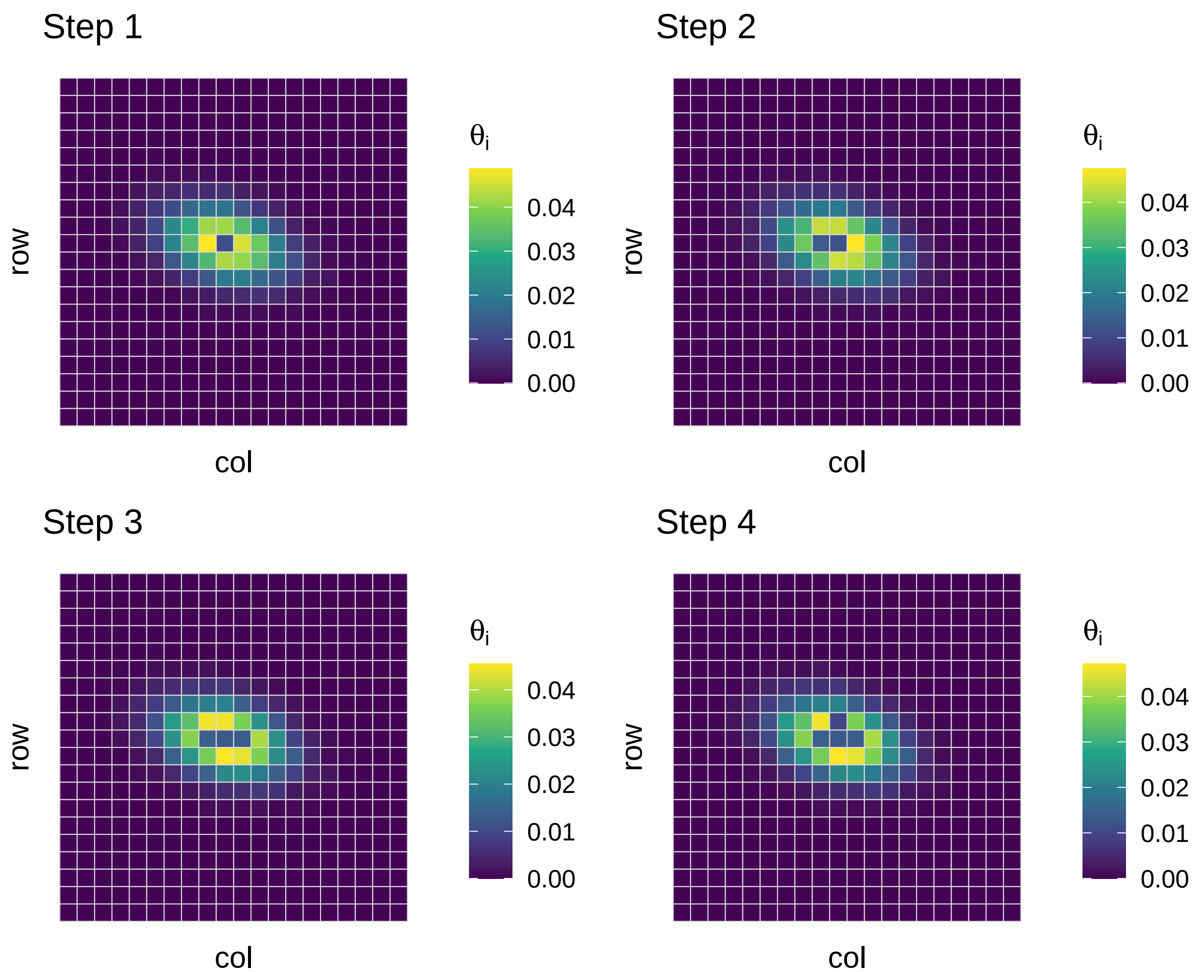

Figure 5: Posterior probability heatmaps for the first four search steps

The heatmaps show the evolution of the posterior probability θ_{ij} over the first four search steps. Initially, the probability mass is relatively distributed throughout the 20×20 grid, reflecting substantial uncertainty about the location of the fisherman under the prior distribution. Higher probabilities are concentrated around the central region of the grid due to the expert-informed prior.

As the search progresses, unsuccessful searches lead to Bayesian updates that reduce the probability in the searched cells. These probabilities are decreased according to the probability of missing the fisherman given his presence in that cell. The removed probability mass is redistributed across the remaining grid, leading to a gradual reshaping of the posterior surface.

Over successive steps, the distribution becomes more concentrated as regions inconsistent with observed search outcomes are down-weighted. This results in a clearer localization of probability mass, with low-probability regions expanding in areas that have been searched without detection.

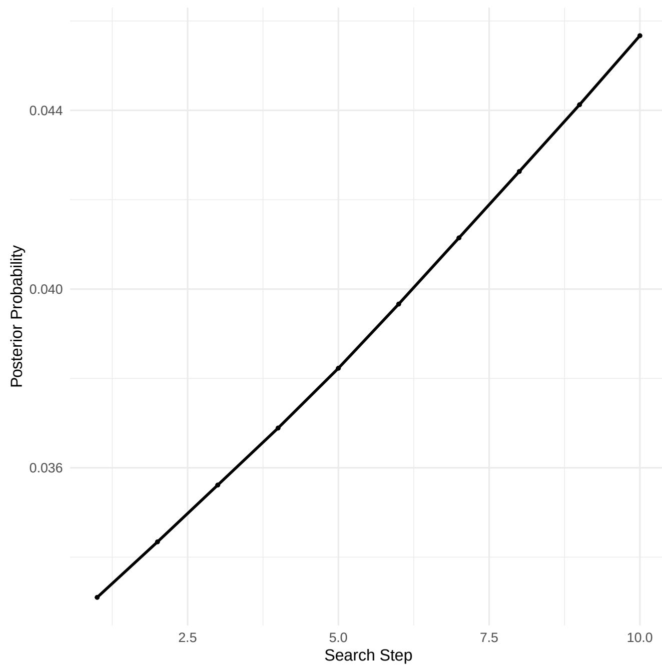

Figure 6: Posterior probabilities at different search steps

The plot shows how the probability of the fisherman being in the true cell changes over the search process. At the start, the probability is low as there is still a lot of uncertainty about where the fisherman could be.

As the search continues, the probability goes up. This happens because when other cells are searched and the fisherman is not found, the probabilities get updated and redistributed across the grid. So even if the true cell is not directly searched, its probability can still change depending on what is happening in the rest of the grid.

Over time, the uncertainty reduces as more cells are ruled out. This causes the probability of the true cell to generally become more focused as the algorithm learns more about where the fisherman is not.

In the end, once the fisherman is detected, the probability jumps up sharply, showing that the model is now very confident about the correct location.

c Comparison of Bayesian and Naive simulation

The number of hours required for the Bayesian search approach is 27.37 hours on average, while the naive simulation method takes 182.34 hours on average for the same Bayesian search problem.

This shows a big difference in performance. The Bayesian approach is much more efficient because it uses prior information and updates its understanding as it searches, which helps it focus on more promising areas instead of searching randomly. On the other hand, the naive method doesn't use any structure or learning, so it spends a lot more time exploring unnecessary parts of the search space.

Overall, the naive method takes about 6.7 times longer than the Bayesian method. This clearly shows that the Bayesian search approach is much faster and more efficient for this problem.

3 A Twist on Linear Regression

a Deriving conditional posterior distributions

First derivation

$$p(\beta \mid \tau_1, \tau_2, X, Y) \propto p(Y \mid X, \beta, \tau_1, \tau_2)\, p(\beta)$$

$$p(Y \mid X, \beta, \tau_1, \tau_2) \propto \exp\left\{-\frac{1}{2}\left[\tau_1 (Y_1 - X_1\beta)^T (Y_1 - X_1\beta) + \tau_2 (Y_2 - X_2\beta)^T (Y_2 - X_2\beta)\right]\right\}$$

$$p(\beta \mid \tau_1, \tau_2, X, Y) \propto \exp\left\{-\frac{1}{2}\left[\beta^T T_0^{-1}\beta + \tau_1 (Y_1 - X_1\beta)^T (Y_1 - X_1\beta) + \tau_2 (Y_2 - X_2\beta)^T (Y_2 - X_2\beta)\right]\right\}$$

$$\propto \exp\left\{-\frac{1}{2}\left[\beta^T \left(\tau_1 X_1^T X_1 + \tau_2 X_2^T X_2 + T_0^{-1}\right)\beta - 2\beta^T \left(\tau_1 X_1^T Y_1 + \tau_2 X_2^T Y_2\right)\right]\right\}$$

The conditional posterior is:

$$\beta \mid \tau_1, \tau_2, X, Y \sim \mathcal{N}(\mu_n, \Sigma_n)$$

where

$$\Sigma_n^{-1} = \tau_1 X_1^T X_1 + \tau_2 X_2^T X_2 + T_0^{-1}$$

$$\mu_n = \Sigma_n \left(\tau_1 X_1^T Y_1 + \tau_2 X_2^T Y_2\right)$$

Second derivation

$$p(\tau_1 \mid \beta, \tau_2, X, Y) \propto p(Y_1 \mid X_1, \beta, \tau_1)\, p(\tau_1)$$

$$p(\tau_1) \propto \tau_1^{a-1} \exp(-b\,\tau_1)$$

$$p(Y_1 \mid X_1, \beta, \tau_1) \propto \tau_1^{n_1/2} \exp\left\{-\frac{\tau_1}{2}(Y_1 - X_1\beta)^T (Y_1 - X_1\beta)\right\}$$

$$= \tau_1^{n_1/2} \exp\left\{-\frac{\tau_1}{2}\sum_{i=1}^{n_1}(y_{1i} - x_{1i}^T\beta)^2\right\}$$

$$p(\tau_1 \mid \beta, \tau_2, X, Y) \propto \tau_1^{a-1} \exp(-b\,\tau_1) \times \tau_1^{n_1/2} \exp\left\{-\frac{\tau_1}{2}\sum_{i=1}^{n_1}(y_{1i} - x_{1i}^T\beta)^2\right\}$$

$$\propto \tau_1^{a+n_1/2-1} \exp\left\{-\tau_1\left[b + \frac{1}{2}\sum_{i=1}^{n_1}(y_{1i} - x_{1i}^T\beta)^2\right]\right\}$$

The conditional posterior is:

$$\tau_1 \mid \beta, \tau_2, X, Y \sim \text{Gamma}\left(a + \frac{n_1}{2},\ b + \frac{1}{2}\sum_{i=1}^{n_1}(y_{1i} - x_{1i}^T\beta)^2\right)$$

Third derivation

$$
\begin{aligned}
p(\tau_2 \mid \tau_1, \beta, X, Y) &\propto p(Y_2 \mid X_2, \beta, \tau_2)\, p(\tau_2 \mid \tau_1) \\
p(\tau_2 \mid \tau_1) &\propto \tau_2^{a-1} \exp(-b\,\tau_2)\, \mathbb{I}(0 < \tau_2 < \tau_1) \\
p(Y_2 \mid X_2, \beta, \tau_2) &\propto \tau_2^{n_2/2} \exp\left\{-\frac{\tau_2}{2}(Y_2 - X_2\beta)^T(Y_2 - X_2\beta)\right\} \\
p(\tau_2 \mid \tau_1, \beta, X, Y) &\propto \tau_2^{a+n_2/2-1} \exp\left\{-\tau_2\left[b + \frac{1}{2}(Y_2 - X_2\beta)^T(Y_2 - X_2\beta)\right]\right\} \mathbb{I}(0 < \tau_2 < \tau_1)
\end{aligned}
$$

The conditional posterior is:

$$
\tau_2 \mid \tau_1, \beta, X, Y \sim \text{Gamma}\left(a + \frac{n_2}{2},\ b + \frac{1}{2}\sum_{i=1}^{n_2}(y_{2i} - x_{2i}^T\beta)^2\right) \mathbb{I}(0 < \tau_2 < \tau_1)
$$

b Code for samples

The joint posterior is:

$$
\begin{aligned}
p(\beta, \tau_1, \tau_2 \mid X, Y) &\propto p(Y \mid X, \beta, \tau_1, \tau_2)\, p(\beta)\, p(\tau_1, \tau_2) \\
p(Y \mid X, \beta, \tau_1, \tau_2) &\propto \tau_1^{n_1/2} \tau_2^{(n-n_1)/2} \exp\left\{-\frac{1}{2}\left[\tau_1(Y_1 - X_1\beta)^T(Y_1 - X_1\beta)\right.\right. \\
&\qquad \left.\left. + \tau_2(Y_2 - X_2\beta)^T(Y_2 - X_2\beta)\right]\right\} \\
p(\beta) &\propto \exp\left\{-\frac{1}{2}\beta^T T_0^{-1}\beta\right\} \\
p(\tau_1, \tau_2) &\propto \tau_1^{a-1}\tau_2^{a-1} \exp\left\{-b(\tau_1 + \tau_2)\right\} \mathbb{I}(0 < \tau_2 < \tau_1) \\
p(\beta, \tau_1, \tau_2 \mid X, Y) &\propto \tau_1^{a+n_1/2-1} \tau_2^{a+(n-n_1)/2-1} \\
&\quad \times \exp\left\{-\frac{1}{2}\beta^T T_0^{-1}\beta - \frac{\tau_1}{2}(Y_1 - X_1\beta)^T(Y_1 - X_1\beta)\right. \\
&\qquad \left. - \frac{\tau_2}{2}(Y_2 - X_2\beta)^T(Y_2 - X_2\beta) - b(\tau_1 + \tau_2)\right\} \\
&\quad \times \mathbb{I}(0 < \tau_2 < \tau_1)
\end{aligned}
$$

Creating Gibbs function

```
# Contaminate the data with outliers
set.seed(12)
n1 <-350 # Number of 'standard' observations
n2 <-n- n1 # Number of outliers
outlier.idx <-(n1 + 1):n # Index set I2
# Inflate the response for the outlier observations
Y.contaminated <-Y
Y.contaminated[outlier.idx] <-Y[outlier.idx] +
  rnorm(n2, mean = 0, sd = 15)

X1 <- X[1:n1,]
X2 <- X[(n1+1):n,]
y1 <- Y.contaminated[1:n1,]
y2 <- Y.contaminated[outlier.idx]

a1 <- 0.1
b1 <- 0.1
a2 <- 0.1
b2 <- 0.1
T_0 <- diag(0.001, ncol(X1))

iterations <- 70000
betas <- matrix(0, iterations, ncol(X1))
tau1s <- tau2s <- numeric(iterations)

tau1 <- 1
tau2 <- 0.5

for (i in 1:iterations) {

  post_variance <- solve(tau1 * crossprod(X1) + tau2 * crossprod(X2) + solve(
      T_0))
  mu <- post_variance %*% (tau1 * t(X1) %*% y1 + tau2 * t(X2) %*% y2)
  beta <- mvrnorm(1, mu, post_variance)

  rss1 <- sum((y1 - X1 %*% beta)^2)
  tau1 <- rgamma(1, a1 + n1/2, b1 + 0.5*rss1)

  rss2 <- sum((y2 - X2 %*% beta)^2)
  repeat {tau2_temp <- rgamma(1, shape = a2 + n2/2, rate = b2 + 0.5*rss2)
  if (tau2_temp <- tau1) break}
  tau2 <- tau2_temp

  betas[i,] <- beta
  tau1s[i] <- tau1
  tau2s[i] <- tau2

}

betas <- betas[(0.2 * iterations + 1):iterations,]
tau1s <- tau1s[(0.2 * iterations + 1):iterations]
tau2s <- tau2s[(0.2 * iterations + 1):iterations]
```

c Trace Plots

Betas

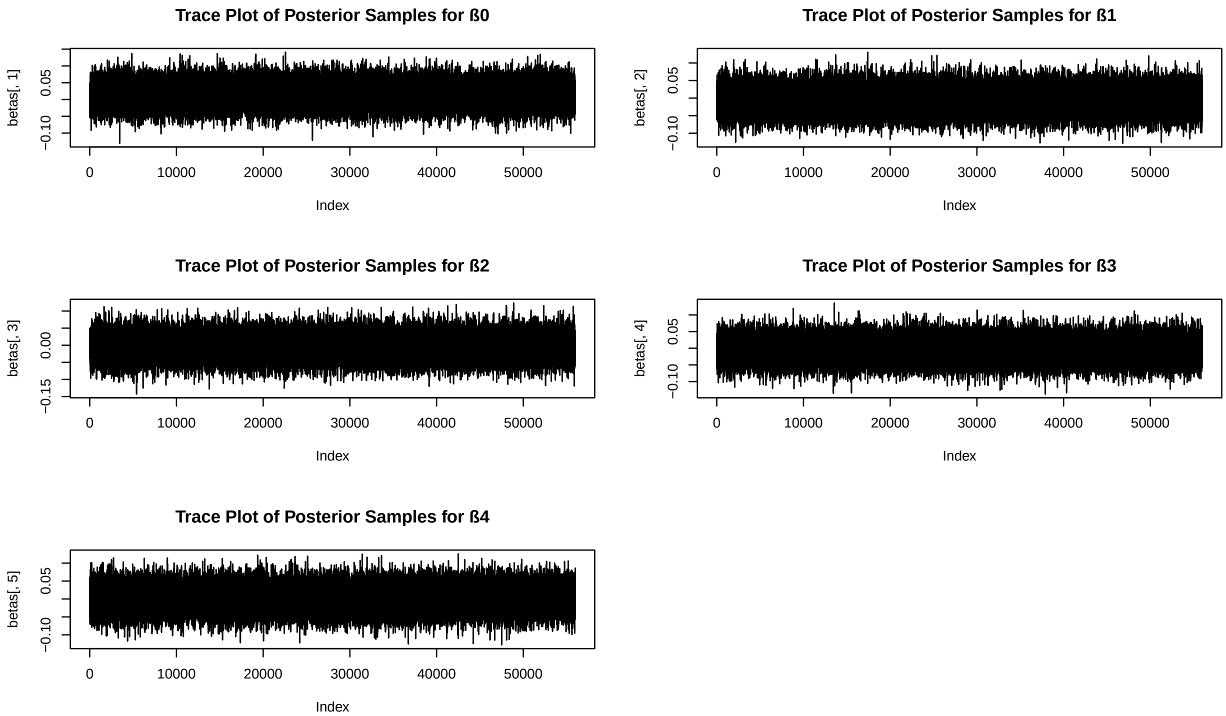

Figure 7: Posterior densities of β_i from informative prior

Taus

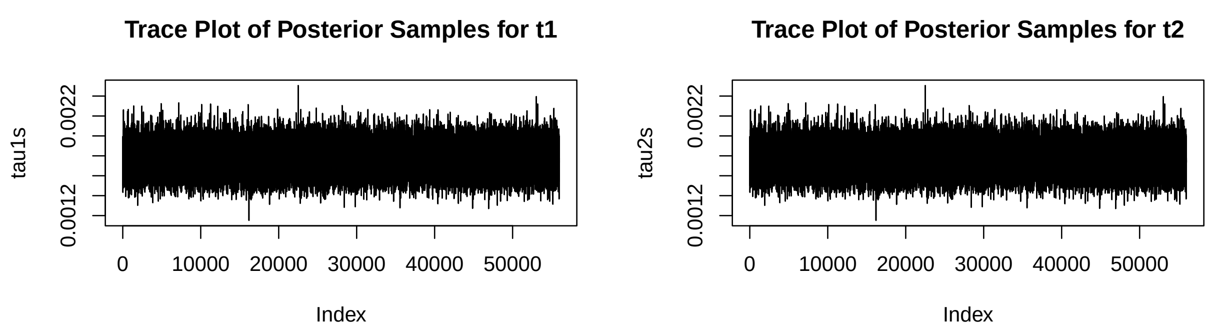

Figure 8: Posterior densities of β_i from informative prior

The trace plots for $\beta_i' s, \tau_1, \tau_2$ show the chains fluctuating around a stable mean with no visible trends over the 50000 iterations (index). This suggests that the Gibbs sampler has reached a stationary distribution and achieved convergence.

Comparing samples

Comparing posterior samples shows all regression coefficients ($\beta_i's$) shifted substantially toward zero relative to those in Question 1, with β_1 moving from 23.386 to 0.016. Posterior standard deviations decreased by a factor of 7-22 across parameters, from 0.22 to 0.71 in the first question to approximately 0.032 in the current model, indicating much greater posterior precision.